$\nabla \times \vec{F}$ in spherical coordinates

Yifan Liu

Department of Physics and Astronomy,
University of California, Riverside, California 92521, USA

E-mail: yliu545@ucr.edu

Abstract: In this note, we present a brief introduction to the calculation of $\nabla \times \vec{F}$ in spherical coordinates. This includes an the minimal mathematical preparation to the fundamental concepts of differential forms and a discussion on Stokes' theorem, which offers a straightforward approach to calculating $\nabla \times \vec{F}$.

Contents

1 Differential form **1**

2 Exterior derivatives of forms **2**

3 Integral of forms and Stokes' theorem **3**

3.1 Green's theorem in 2-D 4

3.2 Stokes' theorem in 3-D 4

3.3 Gauss' theorem in 3-D 4

4 $\nabla \times \vec{F}$ in spherical coordinates **4**

1 Differential form

Mathematically, a differential form is a special type of tensor (totally anti-symmetric tensor) defined on a manifold (geometric object). It is a powerful tool in the sense that it can be used to define "integrand" over curves, surfaces, and higher-dimensional manifolds.

For simplicity we consider 3-D space with differentials dx, dy, dz defined in Cartesian coordinates. Here these differentials are not just infinitesimals which describes the change in the coordinates, but are to be treated as mathematical objects which can be used to define "basis" in the space of differential forms. For example, given functions $\omega_x, \omega_y, \omega_z$ in 3-D space , we can define a linear combination of differentials dx, dy, dz as:

$$\omega_1 = \omega_x(x, y, z)dx + \omega_y(x, y, z)dy + \omega_z(x, y, z)dz, \tag{1.1}$$

which is called "**one-from**" and can be related to the integrand of a line integral. Similarly, one can also define a " **two-from**" of the type

$$\omega_2 = \omega_{xy}(x, y, z)dx \wedge dy + \omega_{yz}(x, y, z)dy \wedge dz + \omega_{zx}(x, y, z)dz \wedge dx \tag{1.2}$$

which appears as the integrand in the surface integrals. Lastly, we can define a " **three-from**" which is related to the integrand of a volume integral,

$$\omega_3 = \omega_{xyz}(x, y, z)dx \wedge dy \wedge dz. \tag{1.3}$$

The symbol $\wedge$ is called "**wedge product/exterior product**". It is a totally anti-symmetric tensor product in the sense that permuting neighboring differentials gives rise to an overall minus sign:

$$dx \wedge dy = -dy \wedge dx, \qquad dx \wedge dz = -dz \wedge dx, \qquad dz \wedge dy = -dy \wedge dz. \tag{1.4}$$

Because of the anti-symmetry, the wedge product of the same differential is always zero as $dx \wedge dx = -dx \wedge dx \to dx \wedge dx = 0$.

Exercise:

- Simplify $\omega = (2dx + dy + 3dz) \wedge (2dx + dy + dz)$
- Simplify $\omega = (dx + 2dy + dz) \wedge (2dx + 4dy + dz) \wedge (dx + dy + dz)$
- Let $A = A_x dx + A_y dy + A_z dz$, $B = B_x dx + B_y dy + B_z dz$ and $C = C_x dx + C_y dy + C_z dz$, find $A \wedge B$ and $A \wedge B \wedge C$. One can see the connection between wedge product and cross product through this example.

2 Exterior derivatives of forms

We now consider the properties of differential forms under differentiation. The corresponding operation is called **exterior derivative** of forms. To see how exterior derivative acts on differential forms, we first consider a zero-form $\omega_0 = f(x, y, z)$ which is also a scalar function. The exterior derivative of zero-form is the differential of a scalar function,

$$d\omega_0 = \partial_x f dx + \partial_y f dy + \partial_z f dz. \tag{2.1}$$

For an one-form with only dx-component $\omega_1 = \omega_x dx$ the corresponding exterior derivative is given by

$$d\omega_1 = (\partial_x \omega_x)\underbrace{dx \wedge dx}_{0} + (\partial_y \omega_x) dy \wedge dx + (\partial_z \omega_x) dz \wedge dx. \tag{2.2}$$

Similarly, for a two-form with only $dxdy$-component $\omega_2 = \omega_{xy} dx \wedge dy$ the corresponding exterior derivative is given by

$$\begin{aligned} d\omega_2 &= (\partial_x \omega_{xy}) dx \wedge dx \wedge dy + (\partial_y \omega_{xy}) dy \wedge dx \wedge dy + (\partial_z \omega_{xy}) dz \wedge dx \wedge dy \\ &= (\partial_z \omega_{xy}) dx \wedge dy \wedge dz \end{aligned} \tag{2.3}$$

where the first two terms vanish due to the anti-symmetry of the wedge product.

From these examples we conclude that the exterior derivative effectively acts as the operation

$$d \sim (\partial_x)(dx\wedge) + (\partial_y)(dy\wedge) + (\partial_z)(dz\wedge), \tag{2.4}$$

where derivatives $\partial_x, \partial_y, \partial_z$ act on the components of forms and $(dx\wedge), (dy\wedge), (dz\wedge)$ act on the basis.

Exercise:

Given the following differential forms,

$$\begin{aligned} \omega_0 &= \omega_0(x, y, z) \\ \omega_1 &= \omega_x(x, y, z) dx + \omega_y(x, y, z) dy + \omega_z(x, y, z) dz \\ \omega_2 &= \omega_z(x, y, z) dx \wedge dy + \omega_x(x, y, z) dy \wedge dz + \omega_y(x, y, z) dz \wedge dx, \end{aligned}$$

calculate the corresponding exterior derivative and show that

$$
\begin{aligned}
d\omega_0 &= (\nabla\omega_0)_x dx + (\nabla\omega_0)_y dy + (\nabla\omega_0)_z dz \\
d\omega_1 &= (\nabla\times\vec{\omega})_x dy\wedge dz + (\nabla\times\vec{\omega})_y dz\wedge dx + (\nabla\times\vec{\omega})_z dx\wedge dy \\
d\omega_2 &= (\nabla\cdot\vec{\omega}) dx\wedge dy\wedge dy,
\end{aligned}
$$

where $\nabla\times\vec{\omega}$, $\nabla\cdot\vec{\omega}$ are the curl and divergence of the vector $\vec{\omega} = (\omega_x, \omega_y, \omega_z)$, respectively. This example shows that the notions of gradient, curl and divergence can be unified using the notions of differential forms and exterior derivatives.

3 Integral of forms and Stokes' theorem

We now define the integrals of differential forms in a way that is compatible with the usual notion of integration. For example, integral of a 1-form in 2-dimensional space over a curve C can be defined as

$$\int_C \omega = \int_C \left[\omega_x(x,y,z)dx + \omega_y(x,y,z)dy\right], \tag{3.1}$$

which is in the form of a line integral and thus can be evaluated using the usual methods. Similarly, the integral of a 2-form ω in 2-dimensional space over a surface S is defined as

$$\int_S \omega = \int_C \omega_{xy}(x,y,z)dx\wedge dy, \tag{3.2}$$

where $dx\wedge dy$ can be interpreted as the area element spanned by the displacements dx and dy. Therefore, the usual notion of integral can be used if we identify $dx\wedge dy \sim dxdy$, and $dy\wedge dx \sim -dxdy$.

We now introduce a foundational principle that bridges the integral of a form over a boundary with the integral of its exterior derivative over the domain it encloses–**Stokes' theorem**. This theorem generalizes various results from classical calculus like the Fundamental Theorem of Calculus and Green's, Gauss's, and Stokes' theorems in vector calculus. To illustrate the idea, we need to first introduce several ingredients:

1. Ω is simple-connected (no holes[1]) p-dimensional region;

2. $\partial\Omega$ is the $p-1$-dimensional boundary of Ω;

3. ω is a $p-1$-form defined on Ω and $\partial\Omega$ and the exterior derivative $d\omega$ is a p-form defined on Ω and $\partial\Omega$;

then Stokes' theorem says

$$\boxed{\int_{\partial\Omega}\omega = \int_\Omega d\omega \qquad (\textbf{Stokes' theorem}).} \tag{3.3}$$

[1]For example, a torus is not simple connected as it contains hole(s).

3.1 Green's theorem in 2-D

In the first example, we consider how the Stokes's theorem of differential forms reproduce the Green's theorem in two dimensions. Let $\omega = P(x,y)dx + Q(x,y)dy$ be an one-from in 2-D space. Then its exterior derivative is given by

$$d\omega = \partial_y P(x,y)dy \wedge dx + \partial_x Q(x,y)dx \wedge dy = [\partial_x Q(x,y) - \partial_y P(x,y)]dx \wedge dy. \tag{3.4}$$

Then Stokes' theorem can be applied to a closed surface S with boundary C,

$$\int_S [\partial_x Q(x,y) - \partial_y P(x,y)]dx \wedge dy = \int_C (P(x,y)dx + Q(x,y)dy). \tag{3.5}$$

This is exactly the Green's theorem in 2-D if we interpret $dx \wedge dy$ as the area element $dS = dxdy$.

3.2 Stokes' theorem in 3-D

Given an one-form $\omega = \omega_x dx + \omega_y dy + \omega_z dz = \vec{\omega} \cdot d\vec{r}$, we can use the result in the exercise of section 2 and write

$$d\omega = (\nabla \times \vec{\omega})_x dy \wedge dz + (\nabla \times \vec{\omega})_y dz \wedge dx + (\nabla \times \vec{\omega})_z dx \wedge dy \tag{3.6}$$

Then we apply Stokes' theorem to a surface S with boundary C and replace $dy \wedge dz \to dA_x$, $dz \wedge dx \to dA_y$, and $dx \wedge dy \to dA_z$, then

$$\int_S (\nabla \times \vec{\omega}) \cdot d\vec{A} = \int_C \vec{\omega} \cdot d\vec{r}, \tag{3.7}$$

which is just the Stokes' theorem in 3-D.

3.3 Gauss' theorem in 3-D

Given a 2-form $\omega = \omega_z(x,y,z)dx \wedge dy + \omega_x(x,y,z)dy \wedge dz + \omega_y(x,y,z)dz \wedge dx$, we can write integral of $d\omega$ over a volume V with boundary ∂V using the generalized Stokes' theorem and the exercise in section 2,

$$\int_V d\omega = \int_V \nabla \cdot \vec{\omega} dx \wedge dy \wedge dz = \int_V \nabla \cdot \vec{\omega} dV = \int_{\partial\Omega} \vec{\omega} \cdot d\vec{A} \tag{3.8}$$

where $dx \wedge dy \wedge dz \to dV$, $dy \wedge dz \to dA_x$, $dz \wedge dx \to dA_y$, and $dx \wedge dy \to dA_z$.

4 $\nabla \times \vec{F}$ in spherical coordinates

In this section, we applied the Stokes' theorem to an one-from in 3-dimensional space and derive curl of a vector field in spherical coordinates.

We parametrize spherical coordinates by (r, θ, ϕ) with unit vectors $(\hat{r}, \hat{\theta}, \hat{\phi})$. The line elements along these directions are given by $d\vec{r} = (dr, rd\theta, r\sin\theta d\phi)$, and the area elements are $d\vec{A} = (dA_r, dA_\theta, dA_\phi) = (r^2 \sin\theta d\theta d\phi, r\sin\theta dr d\phi, r dr d\theta)$.

Consider an one-form $F = \vec{F} \cdot d\vec{r}$. The line integral in spherical coordinates on a closed contour C is

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \left(F_r dr + F_\theta r d\theta + F_\phi r \sin\theta d\phi \right). \tag{4.1}$$

Then Stokes' theorem leads to

$$\begin{aligned}
\int_C \vec{F} \cdot d\vec{r} = \int_C \Bigg(& \frac{\partial F_r}{\partial \phi} d\phi \wedge dr + \frac{\partial F_r}{\partial \theta} d\theta \wedge dr + \frac{\partial (F_\theta r)}{\partial r} dr \wedge d\theta + \frac{\partial (F_\theta r)}{\partial \phi} d\phi \wedge d\theta \\
& + \frac{\partial (F_\phi r \sin\theta)}{\partial r} dr \wedge d\phi + \frac{\partial (F_\phi r \sin\theta)}{\partial \theta} d\theta \wedge d\phi \Bigg) \\
= \int_C \Bigg[& \frac{1}{r \sin\theta} \left(\frac{\partial F_r}{\partial \phi} - \frac{\partial (F_\phi r \sin\theta)}{\partial r} \right) \underbrace{(r \sin\theta d\phi) \wedge dr}_{dA_\theta} \\
& + \frac{1}{r^2 \sin\theta} \left(\frac{\partial (F_\phi r \sin\theta)}{\partial \theta} - \frac{\partial (F_\theta r)}{\partial \phi} \right) \underbrace{(r d\theta) \wedge (r \sin\theta d\phi)}_{dA_r} \\
& + \frac{1}{r} \left(\frac{\partial (F_\theta r)}{\partial r} - \frac{\partial F_r}{\partial \theta} \right) \underbrace{dr \wedge (r d\theta)}_{dA_\phi} \Bigg] \equiv \int_S (\nabla \times \vec{F}) \cdot d\vec{A}.
\end{aligned} \tag{4.2}$$

Therefore,

$$(\nabla \times \vec{F})_r = \frac{1}{r \sin\theta} \left(\frac{\partial (F_\phi \sin\theta)}{\partial \theta} - \frac{\partial F_\theta}{\partial \phi} \right) \tag{4.3}$$

$$(\nabla \times \vec{F})_\theta = \frac{1}{r} \left(\frac{1}{\sin\theta} \frac{\partial F_r}{\partial \phi} - \frac{\partial (F_\phi r)}{\partial r} \right) \tag{4.4}$$

$$(\nabla \times \vec{F})_\phi = \frac{1}{r} \left(\frac{\partial (F_\theta r)}{\partial r} - \frac{\partial F_r}{\partial \theta} \right). \tag{4.5}$$